

#EcoXchange

##Smart Circular Waste Management & Sustainable Marketplace Platform

###“Transforming Waste into Value through Smart Technology.”

Document Type	Product Requirements Document
Product Category	Circular Waste Management, Sustainability, Marketplace, SaaS
Platform Type	Full-Stack MERN Web Application
Primary Users	Citizens, Delivery Agents, Supervisors, Recyclers, Administrators
Design Theme	Modern SaaS, Eco-Friendly, Minimal, Data-Driven
Color Palette	Forest Green, Deep Teal, White, Light Gray, Soft Blue

##1.Product Vision

EcoXchange is a smart circular waste management and sustainable marketplace platform designed to digitize household waste segregation, pickup verification, recycling operations, and eco-commerce.

The platform connects citizens, delivery agents, supervisors, recyclers, and administrators into one transparent ecosystem where waste becomes measurable value. Through QR-based verification, EcoPoints rewards, real-time dashboards, and a circular marketplace, EcoXchange promotes cleaner communities, responsible consumption, and sustainable business growth.

##2.Brand & Design Direction

###2.1Visual Identity

Design Element	Specification
Primary Color	Forest Green
Secondary Color	Deep Teal
Background Color	White
Surface Color	Light Gray
Accent Color	Soft Blue
Typography Style	Clean SaaS typography with bold headers and readable body text
UI Style	Minimal, card-based, spacious, data-focused
Visual Motifs	Leaves, recycling arrows, QR cards, route maps, analytics graphs
Component Style	Rounded cards, subtle shadows, glassmorphism panels, modern icons

###2.2Interface Personality

-  Sustainable
 -  Data-driven
 -  Secure
 -  Operationally efficient
 -  Commerce-ready
 -  Modular
 -  Responsive
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##3.Project Overview

EcoXchange is a full-stack MERN web application that creates a circular economy ecosystem connecting citizens, delivery agents, supervisors, recyclers, and administrators.

The platform rewards households for proper waste segregation, verifies pickups using QR technology, tracks recycling activities, and provides an eco-friendly marketplace where recycled products can be bought and sold.

The system promotes sustainability, transparency, digital waste management, and community participation through automated workflows, real-time tracking, role-based dashboards, wallet rewards, and marketplace transactions.

##4.Problem Statement

Traditional waste management systems often operate with fragmented processes, limited accountability, and minimal citizen participation.

####Key Problems

Problem	Impact
Improper waste segregation	Higher landfill waste and lower recycling quality
Lack of citizen engagement	Low household participation
No transparent pickup verification	Manual disputes and poor accountability
Poor recycling tracking	Limited visibility after waste collection
Inefficient communication	Delayed pickups and service confusion
No incentive mechanism	Reduced motivation for proper segregation
Limited recycling marketplace	Lower commercial value for recycled materials
Manual record keeping	Operational inefficiency and data loss

####EcoXchange Solution

EcoXchange solves these issues using digital workflows, QR verification, rewards, analytics, role-based management, and a sustainable marketplace.

##5.Target Users

User Type	Primary Role	Core Need
Trial Members	New citizens testing the service	Experience waste pickup and earn trial EcoPoints
Permanent Members	Paid members with QR identity and toolkit	Track pickups, earn rewards, access marketplace
Delivery Agents	Field pickup operators	Scan QR codes, collect waste, upload proof
Supervisors	Operational managers	Assign tasks, verify trials, monitor agents
Recyclers	Processing and resale partners	Manage collected waste and sell recycled products
Platform Administrators	System owners	Manage users, roles, approvals, analytics, campaigns

##6.Business Objectives

- Increase household participation in sustainable waste management
- Improve waste segregation accuracy
- Digitize pickup verification
- Reward eco-friendly behavior
- Support recycling businesses
- Build a sustainable marketplace
- Reduce landfill waste
- Promote circular economy adoption
- Improve logistics efficiency
- Generate measurable environmental impact
- Enable scalable city-level operations
- Create a trusted platform for recycled products

##7.Product Goals

Goal	Outcome
Enable QR-based member verification	Transparent pickup confirmation
Reward responsible behavior	Higher retention and engagement
Support trial-to-paid conversion	Revenue and long-term adoption
Digitize recycler operations	Better processing visibility
Launch circular marketplace	Monetize recycled goods and materials
Provide real-time dashboards	Operational control and performance tracking
Improve pickup reliability	Higher customer satisfaction
Create measurable sustainability metrics	Environmental reporting and credibility

##8.Core Product Modules

###8.1Authentication

####Features

- Email login
- Phone OTP login
- JWT authentication
- Role-based access
- Password reset
- Protected routes
- Session handling
- Secure logout
- Account status validation

####User Roles

Role	Access Level
Trial Member	Limited citizen dashboard
Permanent Member	Full citizen dashboard, marketplace, wallet
Delivery Agent	Pickup dashboard and QR scanner
Supervisor	Operational monitoring and verification
Recycler	Recycler dashboard and marketplace tools
Admin	Full platform control

####Acceptance Criteria

- Users can register and log in securely.
 - OTP login validates phone ownership.
 - JWT tokens protect authenticated routes.
 - Unauthorized users cannot access restricted dashboards.
 - Password reset flow securely updates credentials.
 - Role-based navigation displays only permitted modules.
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##9.Trial Membership

###9.1Overview

Trial Membership allows new users to experience EcoXchange for 5 days before becoming permanent members.

###9.2Trial Features

- 5-day trial access
- Daily waste submission
- Supervisor verification
- EcoPoints earning
- Progress tracking
- Eligibility check
- Trial completion status
- Conversion prompt to permanent membership

###9.3Trial User Flow

Trial Registration

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Daily Waste Submission

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Supervisor Verification

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EcoPoints Allocation

↓

Progress Tracking

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Eligibility Review

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Permanent Membership Offer

###9.4Trial Success Conditions

Condition	Requirement
Trial duration	5 days
Daily activity	Waste submission required
Verification	Supervisor approval required
Reward status	EcoPoints credited after validation
Conversion eligibility	Based on completion and verification history

##10. Permanent Membership

###10.1 Overview

Permanent Membership unlocks the complete EcoXchange citizen experience through a one-time membership fee, QR identity card, toolkit allocation, wallet, rewards, pickup tracking, and marketplace access.

###10.2 Membership Details

Item	Description
Membership Fee	₹300 one-time payment
Identity	QR-based member card
Toolkit	Waste segregation starter kit
Wallet	EcoPoints and reward balance
Marketplace	Buy recycled products and sell recyclables
Tracking	Pickup history and live status
Rewards	Coupons, cashback, discounts, wallet credits
Referral	Invite-based EcoPoints earning

###10.3 Toolkit Includes

Toolkit Item	Quantity	Purpose
Color-coded dustbins	3	Waste segregation
Dustbin covers	100	Clean and organized disposal
QR stickers	100	Pickup and waste bag identification

###10.4 Permanent Member Features

- QR identity card
 - Toolkit allocation
 - Wallet
 - EcoPoints
 - Rewards
 - Pickup tracking
 - Referral system
 - Marketplace access
 - Order history
 - Notifications
 - Campaign participation
 - Impact dashboard
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##11.Marketplace

###11.1Overview

The EcoXchange Marketplace enables the buying and selling of recycled, upcycled, and sustainable products. It also allows recyclers and members to sell approved recyclable materials or eco-friendly goods.

###11.2Marketplace Features

- Buy products
- Sell recyclables
- Product listings
- Cart
- Orders
- Wishlist
- Secure payments
- Order tracking
- Seller dashboard
- Product approval workflow
- Inventory management
- Product categories
- Marketplace discounts using EcoPoints

###11.3Marketplace User Flow

Browse Products

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Add to Cart

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Apply Rewards or Discounts

↓

Secure Payment

↓

Order Confirmation

- ↓
Order Tracking
- ↓
Delivery Completion
- ↓
Review and Sustainability Impact Update

###11.4Marketplace Categories

Category	Examples
Recycled Products	Paper goods, plastic alternatives, décor
Upcycled Goods	Home accessories, bags, furniture
Compost Products	Organic compost, soil enhancers
Reusable Products	Bottles, containers, household essentials
Raw Recyclables	Paper, plastic, metal, glass
Eco Kits	Home segregation and sustainability kits

##12.QR Pickup System

###12.1Overview

The QR Pickup System ensures transparent member verification, pickup confirmation, photo proof, EcoPoints allocation, and downstream recycler processing.

###12.2Pickup Flow

- Delivery Agent Scans Member QR
- ↓
Member Verification
 - ↓
Pickup Confirmation
 - ↓
Photo Proof Upload
 - ↓
EcoPoints Awarded
 - ↓
Waste Sent for Recycler Processing

###12.3QR Verification Requirements

Step	Requirement
QR scan	Validate active membership
Member verification	Confirm user identity and service status
Pickup confirmation	Mark pickup as collected
Photo proof	Upload image evidence
Geo-tagging	Capture pickup location
EcoPoints trigger	Award points after successful confirmation
Recycler handoff	Update processing queue

###12.4Pickup Statuses

Status	Meaning
Scheduled	Pickup assigned
In Progress	Agent started route
Arrived	Agent reached location
Verified	QR scan completed
Collected	Waste collected
Proof Uploaded	Photo proof submitted
Validated	Supervisor confirmed pickup
Processed	Recycler accepted waste
Completed	EcoPoints and records finalized

##13.Delivery Agent Dashboard

###13.1Purpose

The Delivery Agent Dashboard enables field agents to manage daily pickup tasks, scan QR codes, navigate routes, upload proof, and track wallet earnings.

###13.2Features

- Daily tasks
- QR scanner
- Route map
- GPS tracking
- Pickup status
- Photo proof upload

- Wallet
- Pickup history
- Notifications
- Assigned member list
- Issue reporting
- Task completion summary

###13.3Dashboard Cards

Card	Displayed Data
Today's Pickups	Assigned, completed, pending
Route Map	Optimized route and active location
QR Scanner	Member scan and verification status
Proof Upload	Pickup photos and notes
Wallet	Agent earnings and incentives
History	Previous pickups and performance

##14.Supervisor Dashboard

###14.1Purpose

The Supervisor Dashboard supports operational control, agent monitoring, pickup assignment, trial user verification, analytics, live tracking, and reporting.

###14.2Features

- Agent monitoring
- Assign pickups
- Verify trial users
- Analytics
- Live tracking
- Reports
- Route overview
- Issue management
- Pickup validation
- Performance leaderboard
- Trial completion review

###14.3Operational Views

View	Function
Live Agent Map	Track delivery agents in real time
Pickup Queue	Manage scheduled and pending pickups
Trial Verification	Approve or reject trial submissions
Performance Dashboard	Monitor agent completion rates
Reports Center	Generate daily, weekly, monthly reports

##15.Recycler Dashboard

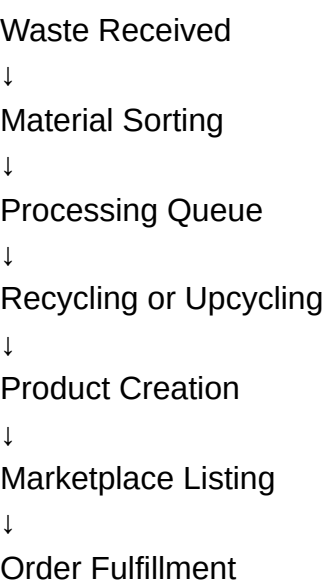
###15.1Purpose

The Recycler Dashboard enables recycling partners to manage incoming waste, product listings, processing queues, payments, marketplace sales, and sustainability metrics.

###15.2Features

- Waste collection
- Product management
- Processing queue
- Marketplace
- Payments
- Sustainability metrics
- Inventory tracking
- Material classification
- Recycled product listing
- Revenue analytics

###15.3Recycler Workflow



###15.4Recycler Metrics

Metric	Purpose
Waste Received	Total material collected
Waste Processed	Recycling throughput
Products Created	Output from recycled materials
Marketplace Sales	Revenue generated
Carbon Saved	Environmental impact
Pending Payments	Financial tracking

##16.Admin Dashboard

###16.1Purpose

The Admin Dashboard provides complete platform governance, user management, role control, marketplace approval, analytics, campaign management, reporting, and system monitoring.

###16.2Features

- User management
- Role management
- Marketplace approval
- Analytics
- Campaigns
- Reports
- Platform monitoring
- Payment overview
- Membership tracking
- QR code management
- Toolkit inventory
- System settings
- Security logs

###16.3Admin Control Panels

Panel	Capability
Users	Create, update, suspend, verify users
Roles	Assign and manage permissions
Marketplace	Approve products and sellers
Pickups	Monitor service operations
Payments	Track memberships, orders, recycler payments
Campaigns	Create sustainability challenges
Analytics	View growth, revenue, and impact
Reports	Export operational and business insights

##17.Reward System

###17.1EcoPoints Overview

EcoPoints are the central engagement mechanism of EcoXchange. Members earn EcoPoints through responsible waste behavior, successful pickups, referrals, and sustainability campaigns.

###17.2EcoPoints Earning Sources

Activity	Reward Trigger
Daily segregation	Verified waste submission
Successful pickups	QR-verified pickup completion
Referrals	New user registration or membership purchase
Campaign participation	Challenge completion
Marketplace activity	Eligible sustainable purchases
Consistent performance	Streak-based rewards

###17.3Reward Types

- Cashback
- Coupons
- Marketplace discounts
- Wallet credits
- Campaign rewards
- Referral bonuses
- Loyalty badges

###17.4Reward Flow

Eco-Friendly Action



Verification



EcoPoints Calculation



Wallet Credit



Reward Redemption



Marketplace or Cashback Usage

##18.System Workflow

###18.1Primary Circular Workflow

Citizen



Waste Segregation



QR Verification



Pickup Scheduled



Delivery Agent Pickup



Supervisor Validation



Recycler Processing



EcoPoints



Marketplace Rewards

###18.2End-to-End Ecosystem Loop

Household Waste



Segregated Collection



Verified Pickup



Recycler Processing



Recycled Products



Marketplace Sales

↓
Consumer Purchase
↓
Circular Economy Impact

##19.Technology Stack

###19.1Frontend

Technology	Purpose
React.js	Frontend application framework
Vite	Fast development and build tooling
JavaScript	Application logic
Tailwind CSS	Utility-first styling
Framer Motion	Animations and micro-interactions
React Query	Server-state management
Zustand	Client-state management

###19.2Backend

Technology	Purpose
Node.js	Backend runtime
Express.js	API framework
MongoDB	NoSQL database
Mongoose	Database modeling
JWT	Authentication and authorization
Socket.IO	Real-time communication

###19.3Cloud Services

Service	Purpose
Cloudinary	Image and proof uploads
Google Maps API	Routes, maps, geolocation
Razorpay	Payments
Firebase OTP	Phone authentication

###19.4Deployment

Platform	Purpose
Vercel	Frontend deployment
Render	Backend deployment
MongoDB Atlas	Cloud database hosting

##20.Database Modules

Module	Core Data
Users	Profiles, roles, authentication, membership status
Wallet	EcoPoints, credits, transactions
Orders	Marketplace purchases and order tracking
Products	Marketplace listings and inventory
Pickups	Schedules, statuses, agent assignments
QR Codes	Member cards, bag stickers, scan records
Toolkits	Kit allocation, inventory, delivery status
Notifications	System alerts and user messages
Campaigns	Sustainability challenges and rewards
Proofs	Pickup photos, timestamps, geolocation
Recycler Payments	Recycler earnings and settlements
Ledger Entries	Financial and reward transaction logs
Membership Purchases	Payment status, plan details, receipts

##21.Security Requirements

###21.1Core Security Features

- JWT authentication
- Password hashing
- Protected routes
- Role-based authorization
- Rate limiting
- Helmet security

- Input validation
- Secure payment handling
- OTP validation
- Environment variable protection
- File upload validation
- API access control

###21.2Security Matrix

Area	Requirement
Authentication	JWT-based login and session validation
Passwords	Hashed and never stored in plain text
Authorization	Access restricted by user role
APIs	Protected endpoints and request validation
Payments	Secure Razorpay transaction flow
Uploads	Validated image uploads via Cloudinary
Forms	Input sanitization and validation
Traffic	Rate limiting for abuse prevention

##22.UI/UX Requirements

###22.1Design Principles

- Modern SaaS interface
- Responsive design
- Light and dark theme
- Interactive dashboard
- Charts and analytics
- Card-based layouts
- Glassmorphism sections
- Minimal design
- Clear navigation
- Accessibility-friendly contrast
- Smooth transitions
- Mobile-first experience

###22.2Core Screens

Screen	Primary Components
Landing Page	Hero, value proposition, impact stats, CTA
Login/Register	Email login, OTP login, password reset
Trial Dashboard	5-day progress, submissions, EcoPoints
Member Dashboard	Wallet, pickups, QR card, rewards
Agent Dashboard	Tasks, scanner, route map, proof upload
Supervisor Dashboard	Live tracking, assignments, verification
Recycler Dashboard	Processing queue, products, payments
Admin Dashboard	Users, roles, analytics, approvals
Marketplace	Products, filters, cart, wishlist
Checkout	Payment, address, rewards application
Reports	Charts, exports, sustainability insights

###22.3Component System

Component	Style Direction
Cards	White or glass panels with soft shadows
Buttons	Forest Green primary, Deep Teal secondary
Badges	EcoPoints, status, role, membership labels
Charts	Soft Blue highlights with green success indicators
Forms	Minimal fields with clear validation states
Tables	Clean, sortable, searchable
Navigation	Sidebar for dashboards, top bar for marketplace
Modals	Focused confirmation and action flows
Maps	Integrated route and pickup visualization

##23.Analytics & Key Metrics

###23.1Platform Metrics

Metric	Description
Users Registered	Total users across all roles
Waste Collected	Total collected waste volume
EcoPoints Earned	Total reward points issued
Pickups Completed	Verified successful pickups
Marketplace Orders	Total orders placed
Carbon Saved	Estimated emissions reduced
Active Members	Members with recent activity
Revenue Generated	Membership and marketplace revenue

###23.2Dashboard Chart Concepts

####Growth Chart

Metric	Visualization
User registrations	Line chart
Trial-to-paid conversion	Funnel chart
Active members	Area chart

####Operations Chart

Metric	Visualization
Daily pickups	Bar chart
Agent performance	Leaderboard
Route completion	Progress ring

####Impact Chart

Metric	Visualization
Waste collected	Stacked bar chart
Carbon saved	Impact counter
Recycling efficiency	Donut chart

####Marketplace Chart

Metric	Visualization
Orders	Line chart
Revenue	Bar chart
Top products	Ranked list

##24.Business Model

###24.1Revenue Streams

Revenue Source	Description
Permanent Membership Fee	₹300 one-time citizen membership
Marketplace Commission	Commission on product sales
Recycler Listings	Featured listings or seller services
Campaign Sponsorships	Partner-sponsored eco campaigns
Analytics Services	Environmental and operational reporting
Bulk Toolkit Sales	Segregation kits and QR sticker packs

###24.2Value Proposition

Stakeholder	Value Delivered
Citizens	Rewards, cleaner homes, transparent pickup
Agents	Structured tasks, verified earnings, route support
Supervisors	Operational control and analytics
Recyclers	Reliable waste supply and marketplace access
Administrators	Scalable platform governance
Communities	Cleaner neighborhoods and lower landfill waste

##25.Feature Priority

Priority	Feature	Release Phase
P0	Authentication and role-based access	MVP
P0	Trial membership workflow	MVP
P0	Permanent membership purchase	MVP
P0	QR pickup verification	MVP
P0	Agent dashboard	MVP
P0	Supervisor verification	MVP
P1	EcoPoints wallet	MVP+
P1	Marketplace buying flow	MVP+
P1	Recycler dashboard	Phase 2
P1	Admin analytics	Phase 2
P2	Referral system	Phase 2
P2	Campaigns	Phase 3
P2	Advanced reports	Phase 3
P3	AI waste classification	Future
P3	IoT smart bins	Future

##26.MVP Scope

###26.1Included in MVP

- User registration and login
- Email and phone OTP authentication
- Role-based dashboards
- Trial membership
- Permanent membership payment
- QR identity card generation
- Toolkit allocation record
- Pickup scheduling
- Delivery agent QR scan flow
- Photo proof upload
- Supervisor verification
- EcoPoints wallet
- Basic marketplace
- Admin user management
- Basic analytics

###26.2Post-MVP Scope

- Advanced marketplace seller tools
 - Recycler financial settlements
 - Campaign management
 - Referral automation
 - Live GPS tracking enhancements
 - Advanced route optimization
 - Carbon credit features
 - AI waste classification
 - IoT smart bin integrations
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##27.User Stories

###27.1Trial Member

User Story	Acceptance Criteria
As a trial member, waste can be submitted daily during the 5-day trial.	Daily submission is recorded with date and status.
As a trial member, EcoPoints can be earned after supervisor verification.	Points are credited only after approval.
As a trial member, trial progress can be tracked.	Dashboard shows completed days and remaining days.

###27.2Permanent Member

User Story	Acceptance Criteria
As a permanent member, membership can be purchased for ₹300.	Payment success activates permanent status.
As a permanent member, a QR identity card is available.	QR card is generated and linked to member ID.
As a permanent member, pickup history can be viewed.	Dashboard displays past and upcoming pickups.
As a permanent member, EcoPoints can be redeemed.	Rewards are applied to eligible actions or orders.

###27.3Delivery Agent

User Story	Acceptance Criteria
As a delivery agent, assigned pickups can be viewed daily.	Task list shows pickup status and location.
As a delivery agent, member QR codes can be scanned.	Scan validates member and updates pickup flow.
As a delivery agent, photo proof can be uploaded.	Proof is linked to pickup record.

###27.4Supervisor

User Story	Acceptance Criteria
As a supervisor, pickups can be assigned to agents.	Agent receives task assignment.
As a supervisor, trial users can be verified.	Trial submission status updates after review.
As a supervisor, agent performance can be monitored.	Dashboard displays completion rate and pending tasks.

###27.5Recycler

User Story	Acceptance Criteria
As a recycler, collected waste can be tracked.	Waste enters processing queue after handoff.
As a recycler, recycled products can be listed.	Product appears after approval.
As a recycler, payments can be tracked.	Dashboard shows pending and completed payments.

###27.6Administrator

User Story	Acceptance Criteria
As an admin, users can be managed.	Admin can view, update, suspend, and assign roles.
As an admin, marketplace products can be approved.	Products require approval before publication.
As an admin, platform analytics can be viewed.	Dashboard displays user, pickup, revenue, and impact metrics.

##28.Notifications

###28.1Notification Types

Trigger	Recipient	Message Purpose
Trial registration	Trial Member	Confirm trial start
Daily reminder	Trial Member	Prompt waste submission
Pickup assigned	Delivery Agent	Notify task assignment
Pickup scheduled	Member	Inform pickup timing
QR scan completed	Member, Supervisor	Confirm verification
Proof uploaded	Supervisor	Request validation
EcoPoints credited	Member	Confirm reward
Marketplace order placed	Buyer, Admin	Confirm order
Product approved	Recycler/Seller	Confirm listing approval
Payment completed	Member/Admin	Confirm transaction

##29.Payment Requirements

###29.1Supported Payment Use Cases

- ₹300 permanent membership purchase
- Marketplace product checkout
- Recycler payment settlements
- Wallet credits
- Reward discount application

###29.2Payment Flow

Select Payment Action

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Create Payment Order

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Razorpay Checkout

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Payment Success Callback

↓

Transaction Verification

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Ledger Entry Created

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Membership/Order/Wallet Updated

###29.3Payment Records

Field	Purpose
Transaction ID	Unique payment reference
User ID	Payment owner
Amount	Payment value
Purpose	Membership, order, wallet, recycler payout
Status	Pending, success, failed, refunded
Timestamp	Audit trail
Ledger Entry	Financial record

##30.QR Code Requirements

###30.1QR Types

QR Type	Purpose
Member QR Card	Permanent member identity verification
Dustbin QR Sticker	Waste bag or bin-level identification
Pickup QR Record	Pickup scan tracking
Toolkit QR	Toolkit allocation and inventory tracking

###30.2QR Data Fields

- QR ID
- Member ID
- Membership status
- Assigned address
- QR type
- Created date
- Active status
- Last scanned date
- Scan history

##31.Data Model Summary

###31.1Users

Field	Description
Name	User full name
Email	Login and communication
Phone	OTP authentication
Role	Access level
Membership Type	Trial or permanent
Address	Pickup location
Wallet ID	EcoPoints wallet reference
Status	Active, suspended, pending

###31.2Pickups

Field	Description
Pickup ID	Unique pickup record
Member ID	Linked household
Agent ID	Assigned delivery agent
Supervisor ID	Validator
Status	Pickup lifecycle state
QR Scan ID	Verification record
Proof ID	Photo proof reference
Location	Pickup coordinates
Scheduled Date	Pickup date
Completed Date	Completion timestamp

###31.3Products

Field	Description
Product ID	Unique listing
Seller ID	Recycler or marketplace seller
Name	Product title
Category	Marketplace category
Price	Selling price
Stock	Available quantity
Images	Product media
Status	Pending, approved, rejected, active
Sustainability Tag	Eco attribute

##32.Platform Architecture

###32.1High-Level Architecture

Client Application

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React.js + Vite Frontend

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Protected API Layer

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Node.js + Express Backend

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MongoDB + Mongoose Database

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Cloud Services

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Payments, OTP, Maps, Storage, Notifications

###32.2Real-Time Capabilities

- Agent live tracking
- Pickup status updates
- Supervisor monitoring
- Notifications
- Marketplace order status
- Wallet updates
- Admin platform monitoring

##33.Reporting Requirements

###33.1Operational Reports

- Daily pickup completion report
- Agent performance report
- Supervisor validation report
- Trial user conversion report
- Waste collection report
- Recycler processing report

###33.2Business Reports

- Membership revenue report
- Marketplace revenue report
- Order volume report
- User growth report
- Reward redemption report
- Active member report

###33.3Impact Reports

- Waste collected
- Recycled material volume
- Carbon saved
- Landfill reduction estimate
- Campaign participation
- Community engagement score

##34.Future Enhancements

Enhancement	Expected Value
AI Waste Classification	Automated identification of waste categories
IoT Smart Bins	Sensor-based fill-level monitoring
Predictive Collection Routes	Optimized pickup routes using data patterns
Carbon Credit Marketplace	Monetization of verified sustainability impact
Government Integration	Municipal-level data exchange and reporting
NGO Partnerships	Community campaigns and awareness programs
Machine Learning Analytics	Forecasting, scoring, and optimization
Mobile Application	Native mobile experience for all roles

##35.Project Impact

###35.1Environmental Impact

- Reduced landfill waste
- Better recycling efficiency
- Lower carbon emissions
- Improved segregation accuracy
- Higher recyclable material recovery
- Measurable sustainability reporting

###35.2Social Impact

- Community engagement
- Citizen awareness
- Cleaner neighborhoods
- Responsible household behavior
- Local employment for pickup agents
- Stronger sustainability culture

###35.3Business Impact

- Digital waste management
- Efficient logistics
- Circular economy ecosystem
- New revenue from marketplace transactions
- Recycler business support
- Scalable urban sustainability model

##36.Success Criteria

Criteria	Success Indicator
✓ Successful QR Pickup Workflow	QR scan, verification, pickup, proof, and reward flow works end-to-end
✓ Trial to Permanent Conversion	Trial members can upgrade through payment
✓ Secure Authentication	Email, OTP, JWT, and role-based access function correctly
✓ Real-Time Tracking	Agent and pickup statuses update reliably
✓ Marketplace Transactions	Products, cart, checkout, and orders operate successfully
✓ Analytics Dashboard	Key metrics display accurately
✓ EcoPoints Rewards	Points are earned, stored, and redeemed
✓ Multi-Role Platform	All dashboards support intended role permissions
✓ Responsive UI	Interface works across desktop, tablet, and mobile
✓ Production Ready	Application is deployed, stable, secure, and scalable

##37.Launch Readiness Checklist

Area	Requirement	Status
Authentication	Email, OTP, JWT, role access	Required
Membership	Trial and permanent flows	Required
Payments	Razorpay integration	Required
QR System	Member QR scanning and verification	Required
Pickups	Scheduling, proof upload, validation	Required
Wallet	EcoPoints credit and redemption	Required
Marketplace	Product listing, cart, order flow	Required
Dashboards	Member, agent, supervisor, recycler, admin	Required
Analytics	Core operational and business metrics	Required
Security	Hashing, validation, protected routes	Required
Deployment	Frontend, backend, database hosting	Required
Responsive Design	Mobile, tablet, desktop support	Required

##38.Product Positioning

EcoXchange positions itself as a premium smart sustainability platform combining waste management, logistics, rewards, and circular commerce into a single digital ecosystem.

###Positioning Statement

EcoXchange helps communities and recycling businesses transform segregated waste into measurable environmental and economic value through QR verification, reward-driven participation, transparent operations, and a sustainable marketplace.

##39.Executive Summary

EcoXchange is a modern circular waste management and marketplace platform built for scalable environmental impact. The platform digitizes the complete lifecycle of household waste, from segregation and QR-based pickup verification to recycler processing and marketplace resale.

With dedicated dashboards for members, delivery agents, supervisors, recyclers, and administrators, EcoXchange enables transparent operations, measurable sustainability outcomes, and incentive-driven citizen engagement.

The product combines smart logistics, EcoPoints rewards, secure payments, marketplace commerce, real-time analytics, and role-based workflows to create a practical, scalable, and production-ready circular economy ecosystem.